Spoof Lehmer Factorizations: Forest Structure, Plus-Seed Chains, and a Closed-Form Finishing Algorithm

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Abstract

We study spoof Lehmer factorizations through the lens of an *extension graph* $\mathcal{G}_k$, building on Hasanalizade's two-factor extension reduction and the Molnar–Singh framework. Three results characterize the global structure of $\mathcal{G}_k$: (i) positive-integer extensions arise exclusively from *plus-seed* states (those with $k\tilde{\varphi}^* = \tilde{\epsilon} + 1$); (ii) the descent formula relating the evaluation of a Lehmer factorization to its descended pair is independent of k; and (iii) $\mathcal{G}_k$ is a **forest** for every $k \geq 2$. We then develop the algorithmic side: three closed-form finishing-feasibility identities (at $s = r-1, r-2, r-3$) replace the deepest levels of bounds-propagation with direct algebraic solves, completing the $r \leq 7$ census (107 factorizations) in minutes where prior code stalled. Tying the two threads together, we prove a *plus-seed chain extension theorem* showing that every plus-seed initiates an infinite chain $F^* \subset F^* \cdot (E+2) \subset \cdots$, of which the Fermat partial products are one chain and $(5, 5, 5, 43, 5377, \ldots)$ is the first known sporadic chain. A coprimality lemma yields a *plus-seed closedness theorem*: from any plus-seed of length s, the length-$(s+2)$ Lehmer extensions number exactly $\tau(M)/2$ where $M = E^2 + E - 1$. This gives closed-form enumeration of two canonical $r = 8$ subtrees (the length-8 extensions of Fermat-6 and sporadic-6 plus-seeds), the first computational data at $r = 8$.

1 Introduction

Molnar and Singh [3] introduced the study of *spoof Lehmer factorizations*: multisets $F = \{x_1, \ldots, x_r\}$ of positive integers satisfying

$$k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) - 1, \qquad k \in \mathbb{Z}_{>0}, \tag{1}$$

where $\tilde{\epsilon}(F) = \prod x_i$ and $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. Hasanalizade [1] proved that if a factorization F satisfies the auxiliary condition $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$ for $\alpha \in \{-1, +1\}$, then the search for two-factor extensions $F \cdot x \cdot y$ satisfying (1) reduces to factoring a single explicit integer (Equation (2) below).

This paper investigates the global structure that the Hasanalizade extension imposes on the solution set, develops a closed-form finishing algorithm for enumeration, and uses both threads to study the family structure that emerges. The main contributions are:

(i) A proof that only $\alpha = +1$ seeds can produce extensions with positive factors (Theorem 2).

(ii) A closed-form descent criterion (Theorem 4) and the observation that it is *independent of k* (Corollary 5).

(iii) A proof that the extension graph $\mathcal{G}_k$ is a forest for every $k \geq 2$ (Theorem 8).

(iv) Three closed-form finishing-feasibility identities at $s = r-1$, $r-2$, $r-3$ (Propositions 13–16) that replace the deepest levels of bounds-propagation with direct algebraic solves.

(v) A complete computational census for $r \leq 7$ across all valid k (Theorem 20), comprising 107 factorizations.

(vi) A plus-seed chain extension theorem (Theorem 22): every plus-seed initiates an infinite chain, of which the Fermat partial products are one instance and $(5, 5, 5, 43, 5377, \ldots)$ is the first known sporadic instance.

(vii) A coprimality lemma (Lemma 26) and resulting plus-seed closedness theorem (Theorem 27): the length-$(s+2)$ Lehmer extensions of any length-s plus-seed number exactly $\tau(M)/2$ for $M = E^2 + E - 1$.

(viii) Closed-form enumeration of two canonical $r = 8$ subtrees, yielding the first concrete $r = 8$ data (Section 10).

2 Preliminaries

We follow the notation of [3] and [1]. A *spoof factorization* is a finite multiset $F = x_1 \cdots x_r$ of positive integers with $x_i \geq 2$, with *spoof evaluation* $\tilde{\epsilon}(F) = \prod x_i$ and *spoof unitary totient* $\tilde{\varphi}^*(F) = \prod(x_i - 1)$. We write $|F| = r$ for the number of factors. A factorization F with $|F| \geq 2$ is *unitary k-Lehmer* if (1) holds.

Definition 1. A factorization F is an *α-seed* (with respect to k) if $k\,\tilde{\varphi}^*(F) = \tilde{\epsilon}(F) + \alpha$, where $\alpha \in \{-1, +1\}$. When $\alpha = +1$, we call F a *plus-seed.*

Hasanalizade's Theorem 3 in [1] states that if F is an α-seed with $N = \tilde{\epsilon}(F)$, then $F \cdot x \cdot y$ is k-Lehmer if and only if

$$(\alpha x - N - \alpha)(\alpha y - N - \alpha) = N^2 + \alpha N - \alpha. \tag{2}$$

We also use the *deficiency function* $g(F) = k\,\tilde{\varphi}^*(F) - \tilde{\epsilon}(F)$, which satisfies the recurrence

$$g(F \cdot a) = (a-1)\,g(F) - \tilde{\epsilon}(F), \qquad g(\emptyset) = k - 1. \tag{3}$$

Lehmer factorizations correspond to $g = -1$, plus-seeds to $g = 1$.

3 Only plus-seeds produce positive extensions

Theorem 2. *Let $k \geq 2$ and let F be a (-1)-seed with $\tilde{\epsilon}(F) = N \geq 1$. Then Equation (2) has no solutions with $x, y \geq 2$.*

Proof. For $\alpha = -1$, Equation (2) becomes $(x + N - 1)(y + N - 1) = N^2 - N + 1$. If $x, y \geq 2$ then the left side is at least $(N+1)^2 = N^2 + 2N + 1 > N^2 - N + 1$ for all $N \geq 1$. □

4 The descent formula is k-independent

Definition 3. Let G be a k-Lehmer factorization. An unordered pair $\{a, b\}$ of factors of G (counted with multiplicity) is a *descended pair* if $F = G \setminus \{a, b\}$ is a plus-seed from which G arises via Equation (2). We call G *primitive* if it admits no descended pair, and *derived* otherwise.

Theorem 4 (Descent formula)**.** *Let $k \geq 2$ and let G be a k-Lehmer factorization with $\tilde{\epsilon}(G) = N$. An unordered pair $\{a, b\}$ of factors of G is a descended pair if and only if*

$$N(a + b - 1) = ab\big((a - 1)(b - 1) + 1\big). \tag{4}$$

Proof. The sub-factorization $F = G \setminus \{a, b\}$ has $\tilde{\epsilon}(F) = N/(ab)$ and $\tilde{\varphi}^*(F) = \tilde{\varphi}^*(G)/\big((a-1)(b-1)\big)$. Since G is k-Lehmer and F is a plus-seed:

$$k\,\tilde{\varphi}^*(F) = \frac{k\,\tilde{\varphi}^*(G)}{(a - 1)(b - 1)} = \frac{N - 1}{(a - 1)(b - 1)}, \qquad k\,\tilde{\varphi}^*(F) = \frac{N}{ab} + 1.$$

Equating:

$$\frac{N - 1}{(a - 1)(b - 1)} = \frac{N}{ab} + 1. \tag{5}$$

Clearing denominators: $(N-1)\,ab = N(a-1)(b-1)+ab(a-1)(b-1)$. Expanding $N(a-1)(b-1) = N(ab - a - b + 1)$ and collecting:

$$N(a + b - 1) = ab\big(ab - a - b + 2\big) = ab\big((a - 1)(b - 1) + 1\big). \qquad \square$$

Corollary 5 (k-independence)**.** *The descent formula* (4) *is independent of k. Equivalently, whether a pair $\{a, b\}$ can be a descended pair is determined entirely by the multiset of factors and the evaluation N, not by the value of k.*

Proof. Equation (5) shows that k cancels in the passage from the k-Lehmer and plus-seed conditions to (4). $\square$

Remark 6. The k-independence is perhaps surprising: the Lehmer condition $k\tilde{\varphi}^* = \tilde{\epsilon} - 1$ and the plus-seed condition $k\tilde{\varphi}^* = \tilde{\epsilon} + 1$ both involve k explicitly, but dividing the first by the second at the level of a descended pair absorbs k into the relationship between $\{a, b\}$ and the remaining factors.

5 The extension graph is a forest

Definition 7. The *extension graph* $\mathcal{G}_k$ is a directed graph whose vertices are the k-Lehmer factorizations with factors ≥ 2. There is an edge $G \to F$ whenever $F = G \setminus \{a, b\}$ for some descended pair $\{a, b\}$.

Theorem 8 (Forest theorem)**.** *For every $k \geq 2$, the extension graph $\mathcal{G}_k$ is a forest: every k-Lehmer factorization with factors ≥ 2 admits at most one descended pair.*

By Corollary 5, the descent formula (4) is k-independent, so the entire proof reduces to showing that no single multiset G with $\tilde{\epsilon}(G) = N$ can have two distinct unordered pairs satisfying (4) with an appropriate divisibility and positivity structure. The proof uses three lemmas, covering the three possible cases for the intersection of two descended pairs.

Lemma 9 (Disjoint case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{c, d\}$ with $\{a, b\} \cap \{c, d\} = \emptyset$.*

Proof. Write $G = \{a, b, c, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$ (where $Q = 1$ if $E = \emptyset$), so $N = abcdQ$. Applying the descent formula (4) to each pair and dividing by ab and cd respectively:

$$cdQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{6}$$
$$abQ\,(c + d - 1) = (c-1)(d-1) + 1. \tag{7}$$

Since $(a-1)(b-1)+1 \leq ab$ for $a, b \geq 2$ (with strict inequality unless one of a, b equals 1), multiplying gives

$$a^2b^2c^2d^2\,Q^2\,(a+b-1)(c+d-1) \leq a^2b^2c^2d^2,$$

whence $Q^2(a + b - 1)(c + d - 1) \leq 1$. Since $a + b - 1 \geq 3$ and $c + d - 1 \geq 3$, we get $Q^2 \leq 1/9 < 1$, so $Q < 1$.

If $E \neq \emptyset$ then $Q \geq 2$, a contradiction. If $E = \emptyset$ then $Q = 1$ and (6)–(7) become $cd(a + b - 1) = (a - 1)(b - 1) + 1$ and $ab(c + d - 1) = (c - 1)(d - 1) + 1$. The first gives $cd < ab/(a + b - 1) \leq ab/3$; the second gives $ab < cd/(c + d - 1) \leq cd/3$. Together: $cd < ab/3 < cd/9$, which is absurd. □

Lemma 10 (Overlapping case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit two descended pairs $\{a, b\}$ and $\{b, d\}$ sharing one element, where $a \neq d$.*

Proof. Write $G = \{a, b, d\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = abdQ$. The descent formula for each pair gives

$$dQ\,(a + b - 1) = (a-1)(b-1) + 1, \tag{8}$$
$$aQ\,(b + d - 1) = (b-1)(d-1) + 1. \tag{9}$$

Bounding: $dQ \leq ab/(a + b - 1)$ and $aQ \leq bd/(b + d - 1)$. Multiplying:

$$adQ^2 \leq \frac{ab^2d}{(a + b - 1)(b + d - 1)}, \qquad \text{so} \qquad Q^2 \leq \frac{b^2}{(a + b - 1)(b + d - 1)}.$$

We claim $(a + b - 1)(b + d - 1) > b^2$ for all $a, b, d \geq 2$. Expanding:

$$\begin{aligned}(a + b - 1)(b + d - 1) - b^2 &= a(b + d - 1) + b(d - 2) + (1 - d)\\ &\geq 2(b + d - 1) + 0 + (1 - d) = 2b + d - 1 \geq 3.\end{aligned}$$

Therefore $Q^2 < 1$, which is impossible if $E \neq \emptyset$ (since then $Q \geq 2$).

If $E = \emptyset$, then $|G| = 3$, $Q = 1$, and $G = \{a, b, d\}$. The seed of $\{a, b\}$ is the single-element factorization (d), and the plus-seed condition $k(d - 1) = d + 1$ gives $d = (k + 1)/(k - 1)$. By symmetry, the seed of $\{b, d\}$ forces $a = (k + 1)/(k - 1)$. We consider three sub-cases:

- $k = 2$: $d = a = 3$, $G = \{3, b, 3\}$, $N = 9b$. The Lehmer condition $2 \cdot 2(b - 1) \cdot 2 = 9b - 1$ gives $8b - 8 = 9b - 1$, hence $b = -7$.
- $k = 3$: $d = a = 2$, $G = \{2, b, 2\}$, $N = 4b$. The Lehmer condition $3 \cdot 1 \cdot (b - 1) \cdot 1 = 4b - 1$ gives $3b - 3 = 4b - 1$, hence $b = -2$.
- $k \geq 4$: $(k + 1)/(k - 1)$ is not an integer (since $k - 1 \geq 3$ and $\gcd(k - 1, k + 1) \mid 2$), so no single-element plus-seed exists and the $|G| = 3$ case is vacuous.

In all cases, no valid factorization arises. □

Lemma 11 (Repeated-element case)**.** *Let G be a k-Lehmer factorization. Then G cannot admit descended pairs $\{v, v\}$ and $\{v, w\}$ with $w \neq v$, where v appears with multiplicity at least two in G.*

Proof. Write $G = \{v, v, w\} \cup E$ with $\tilde{\epsilon}(E) = Q$, so $N = v^2 wQ$. The descent formula for the pair $\{v, v\}$ gives, after dividing by v^2:

$$wQ(2v-1) = (v-1)^2 + 1. \tag{10}$$

The left side is a positive multiple of $2v-1$, so $(2v-1) \mid \big((v-1)^2+1\big)$. We compute modulo $2v-1$: since $2(v-1) \equiv -1 \pmod{2v-1}$,

$$4\big((v-1)^2+1\big) = (2v-2)^2 + 4 \equiv 1 + 4 = 5 \pmod{2v-1}.$$

As $\gcd(4, 2v-1) = 1$, we conclude $(2v-1) \mid 5$, giving $2v-1 \in \{1, 5\}$ and $v \in \{1, 3\}$.

For $v = 3$: Equation (10) gives $5wQ = 5$, so $wQ = 1$, forcing $w = 1$ and $Q = 1$. But factors must be ≥ 2.

For $v = 1$: factors must be ≥ 2.

Neither case yields a valid factorization. □

Proof of Theorem 8. Let G be a k-Lehmer factorization with factors ≥ 2, and suppose for contradiction that $\{a, b\}$ and $\{c, d\}$ are two distinct descended pairs. There are three cases:

- $\{a, b\} \cap \{c, d\} = \emptyset$: impossible by Lemma 9.
- $|\{a, b\} \cap \{c, d\}| = 1$ with distinct values: after relabeling, $\{a, b\}$ and $\{b, d\}$ with $a \neq d$, impossible by Lemma 10.
- $|\{a, b\} \cap \{c, d\}| = 1$ with a repeated element: $\{v, v\}$ and $\{v, w\}$, impossible by Lemma 11.

The only remaining possibility $\{a, b\} = \{c, d\}$ contradicts the assumption that the pairs are distinct. □

Corollary 12. *For any $k \geq 2$, every derived k-Lehmer factorization is produced by a unique plus-seed.*

6 Finishing-feasibility identities

The forest structure (Section 5, Theorem 8) characterizes which factorizations descend from which seeds, but explicit enumeration still requires a search procedure. We now develop three closed-form identities that replace the deepest levels of the bounds-propagation algorithm of [3] with direct algebraic solves. These are the algorithmic backbone of the census in Section 8.

6.1 Finishing-feasibility at $s = r - 1$

Let $F_0 = (x_1, \ldots, x_{r-1})$ be a partial factorization with $\tilde{\epsilon}(F_0) = E$ and $\tilde{\varphi}^*(F_0) = P$. A single factor x_r completes F_0 to a k-Lehmer factorization iff $kP(x_r - 1) = Ex_r - 1$. Solving:

$$x_r = \frac{kP-1}{kP-E}. \tag{11}$$

Proposition 13 ($s = r - 1$ finishing-feasibility)**.** *A partial factorization F_0 of length $r-1$ extends to a k-Lehmer factorization if and only if $kP - E > 0$, $kP - E$ divides $kP - 1$, and the quotient $x_r = (kP-1)/(kP-E)$ satisfies $x_r \geq x_{r-1}$ and the appropriate parity condition. The extension, when it exists, is unique.*

Proof. Let x_r extend F_0 to a k-Lehmer factorization. Then $kP(x_r - 1) = Ex_r - 1$, rearranging to $(kP - E)x_r = kP - 1$. If $kP - E \leq 0$, the left side is non-positive for $x_r \geq 1$ and the right side is positive, a contradiction. Otherwise $x_r = (kP-1)/(kP-E)$ is forced, and integrality requires $kP - E$ to divide $kP - 1$. □

6.2 Finishing-feasibility at $s = r - 2$

Let $F_0 = (x_1, \ldots, x_{r-2})$ with $\tilde{\epsilon}(F_0) = E$ and $\tilde{\varphi}^*(F_0) = P$. We seek a pair (a, b) with $a \le b$ and $a \ge x_{r-2}$ completing F_0 to a k-Lehmer factorization. The defining equation expands to $kP(a-1)(b-1) = Eab - 1$, which rearranges to $(kP - E)\,ab - kP(a+b) + (kP+1) = 0$. Write $A = kP - E$ and $B = kP$. Multiplying by A and completing the product on a and b yields

$$(Aa - B)(Ab - B) \;=\; BE - A. \tag{12}$$

Proposition 14 ($s = r - 2$ finishing-feasibility)**.** *Let F_0 be a partial factorization of length $r - 2$, with $A = kP - E$ and $B = kP$. A pair (a, b) with $a \le b$ and $a \ge x_{r-2}$ extends F_0 to a k-Lehmer factorization if and only if $A > 0$ and there is a divisor pair (d_1, d_2) of $BE - A$ with $d_1 \le d_2$ such that*

$$a = \frac{d_1 + B}{A}, \qquad b = \frac{d_2 + B}{A},$$

are integers satisfying $x_{r-2} \le a \le b$ and the appropriate parity and Lehmer-congruence conditions.

Remark 15 (Relationship to the descent formula). When F_0 is a plus-seed—i.e., $kP = E + 1$, so $A = 1$ and $B = E+1$—Equation (12) becomes $(a-E-1)(b-E-1) = (E+1)E-1 = E^2+E-1 = \Delta_+(E)$, exactly the Hasanalizade descent equation (2). The $s = r - 2$ finishing-feasibility identity therefore subsumes the Hasanalizade descent equation: the descent formula handles the case $A = 1$; Proposition 14 handles arbitrary $A \ge 1$.

6.3 Finishing-feasibility at $s = r - 3$

Let $F_0 = (x_1, \ldots, x_{r-3})$. We seek a triple (a, b, c) with $x_{r-3} \le a \le b \le c$ completing F_0. Substituting $u = Aa - B$, $v = Ab - B$, $w = Ac - B$ in $kP(a-1)(b-1)(c-1) = Eabc - 1$ and multiplying by A^3, we obtain

$$uvw \;-\; EB(u + v + w) \;=\; C_{F_0}, \qquad C_{F_0} := A^2(E-1) + E^2(3A + 2E). \tag{13}$$

For each fixed $u \ge 1$, the remaining two-variable equation multiplies through by u and completes on v, w to

$$(uv - EB)(uw - EB) \;=\; u \cdot C_{F_0} + u^2 EB + (EB)^2 =: M_u. \tag{14}$$

Proposition 16 ($s = r - 3$ finishing-feasibility)**.** *A triple (a, b, c) with $x_{r-3} \le a \le b \le c$ extends F_0 to a k-Lehmer factorization if and only if $A > 0$ and there exist a positive integer u and a divisor pair (D_1, D_2) of M_u satisfying $A \mid u + B$, $u \mid D_1 + EB$, $u \mid D_2 + EB$, $A \mid v + B$, $A \mid w + B$ (where v, w are computed from D_1, D_2), $a \le b \le c$, and the parity/Lehmer-congruence conditions.*

The outer loop over u is bounded: since $v, w \ge u$, we have $uvw \ge u^3$, so $u^3 \le 3EBu + C_{F_0}$, giving $u \le \sqrt{3EB} + O(C_{F_0}^{1/3})$ to leading order. For the case of greatest interest—when F_0 is a plus-seed, where the $s = r - 3$ identity governs $r = s + 3$ extensions—this bound admits a clean closed form:

Proposition 17 ($u_{\max}$ for plus-seeds)**.** *Let F^* be a $k = 2$ plus-seed with $\tilde{\epsilon}(F^*) = E$ (so $A = 1$, $B = E + 1$, $C_{F^*} = 2E^3 + 3E^2 + E - 1$). The $s = r - 3$ finishing-feasibility identity at F^* admits a positive integer solution u only in the range*

$$1 \;\le\; u \;\le\; 2E + 1.$$

Proof. The constraint $u \le v \le w$ together with Equation (13) gives $u^3 \le uvw = EB(u+v+w) + C_{F^*} \le 3EBw + C_{F^*}$ when $w \ge u$; combined with $u \le w$ this yields $u^3 \le 3EBu + C_{F^*}$. At $u = 2(E+1) = 2B$, the polynomial identity

$$(2E+2)^3 - 3E(E+1)(2E+2) - (2E^3 + 3E^2 + E - 1) \;=\; 9E^2 + 17E + 9$$

is positive for all $E \ge 0$. Hence $u^3 > 3EBu + C_{F^*}$ at $u = 2B$, so any productive u must satisfy $u < 2B$, i.e. $u \le 2B - 1 = 2E + 1$. □

Remark 18 (Empirical gap). The bound $u \le 2E+1$ matches the largest real root of $u^3 = 3EBu + C_{F^*}$ up to an $O(1)$ correction; it is essentially the best obtainable from real-valued analysis. Empirically, however, productive u tend to be substantially smaller. For both length-4 plus-seeds appearing in our $r \le 7$ census—the Fermat-4 prefix $(3, 5, 17, 257)$ with $E = 65535$ and the sporadic $(5, 5, 5, 43)$ with $E = 5375$—the maximum observed productive u values are $38557 \approx 0.588 \cdot E$ and $3373 \approx 0.628 \cdot E$ respectively, well below the provable bound $u \le 2E+1$. This suggests the empirical conjecture $u_{\max}^2 \le EB$ for $k = 2$ plus-seeds (equivalently $u \le \sqrt{E(E+1)} \approx E$). Spot-checks on the length-5 $k = 2$ plus-seeds confirm the pattern: sampling u at logarithmically spaced values in the range $(\sqrt{EB}, 2E+1]$ across three length-5 plus-seeds—namely $(5, 7, 7, 7, 133)$ with $E = 228\,095$, $(3, 9, 9, 23, 131)$ with $E = 732\,159$, and $(5, 5, 5, 47, 453)$ with $E = 2\,661\,375$—produced zero productive u beyond $\sqrt{EB}$. However, the divisor-density heuristic does not produce this bound: explicit computation of $\tau(M_u)/u$ at productive u shows values as small as $\sim 10^{-4}$, so the existence of valid divisors is not predicted by uniform-distribution arguments. The gap originates in the integrality requirement on $w = (D_2 + EB)/u$ together with the divisor structure of M_u in the residue class $-EB \pmod{u}$, which does not admit a closed-form bound.

The pattern halts at $j = r - s = 3$: the substitution $u_i = Ax_i - B$ reduces the defining equation to a j-variable homogeneous-degree-j problem; at $j = 2$ it is a hyperbola (exact divisor solution); at $j = 3$ it is a cubic surface, planar only after fixing one variable; at $j \ge 4$ even one-variable-fixed reductions are non-planar.

7 Algorithm

The sharpened bounds-propagation algorithm replaces the last three levels of recursion with direct solves:

Algorithm 1 Sharpened bounds propagation for k-Lehmer enumeration

```
procedure ENUMERATE(F_0, k, r)
    s ← |F_0|
    if s = r − 1 then
        solve (11); emit if integer and valid
    else if s = r − 2 then
        factor BE − A; emit every divisor pair giving valid (a, b)
    else if s = r − 3 then
        for each valid u with u^3 ≤ 3EBu + C_{F_0} do
            factor M_u; emit every divisor pair giving valid (b, c)
        end for
    else
        apply bounds propagation as in [3]
        for each candidate x_{s+1} not pruned by L, U or Lehmer's congruence do
            ENUMERATE(F_0 ∪ {x_{s+1}}, k, r)
        end for
    end if
end procedure
```

Remark 19 (Cost structure). The dominant cost is the integer factorizations at $s = r - 2$ and $s = r - 3$. For $r = 7$ at $k = 2$ odd, the integers to be factored have at most 20 digits, and modern general-purpose factoring routines (Pollard-rho plus ECM as implemented in SymPy [5]) factor each in well under a millisecond. A naive implementation using trial division up to $\sqrt{M}$ is catastrophic here: M reaches $\sim 10^{19}$, so the trial-division loop runs for several minutes per prefix, where the factoring-based implementation finishes in microseconds.

8 The complete $r \leq 7$ census

We implemented Algorithm 1 and ran it for every (r, k) pair with $r \leq 7$, $k \geq 2$, and factors odd and ≥ 3. The relevant k range is finite: any k-Lehmer factorization with minimum factor $a_{\min}$ and length r satisfies $k < (a_{\min}/(a_{\min} - 1))^r$, so for odd factors $k \leq \lfloor (3/2)^r \rfloor$. At $r = 7$ this gives $k \in \{2, 3, \ldots, 17\}$.

Theorem 20 ($r \leq 7$ census)**.** *The complete set of k-Lehmer factorizations with $r \leq 7$ and all factors odd and ≥ 3, across all valid $k \geq 2$, consists of exactly* 107 *factorizations distributed as in Table 1. Only the values $k \in \{2, 4, 5, 8\}$ produce any such factorizations in this range. Of the* 103 *factorizations at $k = 2$,* 50 *are Hasanalizade-descended (their length-$(r - 2)$ prefix is a plus-seed),* 11 *are Lehmer companions of plus-seeds (their length-$(r - 1)$ prefix P^* is a plus-seed and the final factor is $\tilde{\epsilon}(P^*)$),* 41 *are truly primitive (no plus-seed parent of either kind), and one is the trivial* $(3, 3)$*. The full hierarchical listing appears in Appendix A.*

Proof. By exhaustive enumeration using Algorithm 1. The forest property of Theorem 8 guarantees the parent-uniqueness asserted for non-primitive factorizations. □

8.1 Plus-seed chains observed in the census

The forest theorem (Theorem 8) guarantees every non-primitive $k = 2$-Lehmer factorization in the census descends from a unique plus-seed via Hasanalizade two-factor descent. The chain-extension

Table 1: Distribution of odd k-Lehmer factorizations by length r and parameter k, for $2 \leq r \leq 7$. Columns for $k \notin \{2, 4, 5, 8\}$ are omitted (all zeros in this range).

r	$k = 2$	$k = 4$	$k = 5$	$k = 8$	total
2	1	—	—	—	1
3	1	—	—	—	1
4	1	—	1	—	2
5	4	—	—	—	4
6	22	1	—	—	23
7	74	1	—	1	76
total	103	2	1	1	**107**

theorem (Theorem 22) further organizes the plus-seeds themselves into infinite chains rooted at *fresh* plus-seeds (those whose length-$(s-1)$ prefix is not a plus-seed). Combining these, each $k = 2$-Lehmer factorization is attributable to a unique chain (or is truly primitive).

In the $r \leq 7$ census we observe twelve such chains (Table 2): the Fermat chain rooted at $\emptyset$ and eleven sporadic chains. Six of the eleven sporadic chains have their roots at length 6, detected here via inverting the parent relationship from the $r = 7$ census without performing a length-6 plus-seed search. The complete hierarchical listing of every factorization in the census, organized under its parent plus-seed, is given in Appendix A.

Table 2: Plus-seed chains observed in the $r \leq 7$ census ($k = 2$, odd). Each chain is rooted at a *fresh* plus-seed and continues by $x = \tilde{\epsilon} + 2$ (Theorem 22). The *member lengths* column lists the lengths at which the chain has been observed in the census; the *Lehmer descendants* column gives the total number of $k = 2$-Lehmer factorizations descending from any chain member, counting both Hasanalizade-descended and Lehmer-companion children.

Chain	Root	Member lengths	Lehmer descendants
Fermat	$\emptyset$	$1, 2, 3, 4, 5$	24
sporadic	$(5, 5, 5, 43)$	$4, 5$	10
sporadic	$(3, 5, 17, 353, 929)$	5	5
sporadic	$(3, 9, 9, 23, 131)$	5	2
sporadic	$(5, 5, 5, 47, 453)$	5	9
sporadic	$(5, 7, 7, 7, 133)$	5	5
sporadic	$(3, 5, 17, 257, 65729, 22318913)$	6	1
sporadic	$(3, 5, 17, 365, 855, 7234467)$	6	1
sporadic	$(3, 11, 11, 11, 677, 3659)$	6	1
sporadic	$(5, 5, 5, 43, 5413, 786349)$	6	1
sporadic	$(5, 5, 5, 53, 215, 158265)$	6	1
sporadic	$(5, 7, 7, 13, 13, 619)$	6	1

Remark 21 (Length-6 fresh plus-seeds without a length-6 search). The six length-6 sporadic chain roots in Table 2 were not found by an exhaustive length-6 plus-seed search (which is prohibitively expensive at this length). Each appears in our census as the length-$(r-1)$ prefix of an $r = 7$ Lehmer-companion factorization $(P^*, \tilde{\epsilon}(P^*))$, where P^* is the length-6 plus-seed. Inverting this relationship across the census yields the complete length-6 plus-seed inventory *up to the limits of the* $r = 7$

census—namely, every length-6 plus-seed whose Lehmer companion has factors small enough to appear in the $r \leq 7$ search window. Length-6 plus-seeds whose Lehmer companions exceed those bounds (or which produce only length-8 or longer descendants) would not be observed by this method, so the count of 11 sporadic chains is a lower bound on the true count of $r \leq 7$-attributable chains.

8.2 Family structure at $r = 7$, $k = 2$

The 74 factorizations at $r = 7$, $k = 2$ group by length-5 prefix into families. Of the 44 length-5 prefixes appearing, exactly four produce more than one length-7 completion, and *all four are plus-seeds*:

- $(3, 5, 17, 257, 65537) = F^{(5)}$ (Fermat-5): 16 completions.
- $(3, 5, 17, 353, 929)$ (sporadic): 4 completions.
- $(5, 5, 5, 47, 453)$ (sporadic): 8 completions.
- $(5, 7, 7, 7, 133)$ (sporadic): 4 completions.

This corroborates the descent picture: completion multiplicity at length $r - 2$ is governed by $\tau(M)/2$ where $M = E^2 + E - 1$, and multiplicity ≥ 2 requires the prefix to be a plus-seed (so that no divisor of M is excluded by the divisibility-mod-A constraint). At $r = 7$ with $k \neq 2$, exactly two factorizations exist: $(3, 3, 3, 9, 23, 131, 732159)$ at $k = 4$ and $(3, 3, 3, 3, 5, 5, 89)$ at $k = 8$.

9 Plus-seed chains

The plus-seeds in our census do not arise in isolation: they organize themselves into infinite *chains* via a canonical extension operation. The Fermat partial products form one such chain, but they are not the only one. This section develops the chain theory and identifies its first sporadic instance.

9.1 The canonical chain extension

Theorem 22 (Chain extension exists and is unique)**.** *Let F^* be a $k = 2$ plus-seed of length s with $\tilde{\epsilon}(F^*) = E$ and $\tilde{\varphi}^*(F^*) = P$. Define $x := E+2$. Then (F^*, x) is again a $k = 2$ plus-seed. Moreover, x is the unique value of the next factor for which this is the case.*

Proof. For (F^*, x) to be a plus-seed we require $2P(x-1) = Ex+1$, i.e. $x(2P-E) = 2P+1$. Since F^* is itself a plus-seed, $2P-E = 1$ and $2P = E+1$, so the equation simplifies to $x = 2P+1 = E+2$. Substituting $x = E+2$ back, we have $2P(x-1) = (E+1)(E+1) = E^2+2E+1 = E(E+2)+1 = Ex+1$, so (F^*, x) is indeed a plus-seed. □

Lemma 23 (Chain extension is admissible)**.** *The next factor $x = E + 2$ in Theorem 22 is odd, satisfies $x > x_s$ (the last factor of F^*), and respects the Lehmer congruence $x \not\equiv 1 \pmod{a}$ for every prior factor a in F^*.*

Proof. Parity. Each factor of F^* is odd (we restrict to the odd-spoof setting throughout), so E is odd and $E + 2$ is odd.

Ordering. $E = \prod_i x_i \geq x_s$ since each $x_i \geq 1$, and in fact $E + 2 > x_s$ for any non-trivial F^*.

Lehmer congruence. $x-1 = E+1 = 2P$. We need $a \nmid 2P$ for every prior factor a in F^*. Since a is odd, this is equivalent to $a \nmid P = \prod_j (x_j - 1)$. By the Lehmer congruence on F^* itself, $a \nmid (x_j - 1)$

for all j with x_j a factor of F^* (taking j to be the index of a for the trivial case, where $a \nmid (a-1)$ since $a \geq 3$). Hence $a \nmid P$. $\square$

Corollary 24 (Plus-seed chains are infinite)**.** *Every $k = 2$ plus-seed F^* generates an infinite chain*

$$F^* \;\subset\; F^* \cdot (E_0 + 2) \;\subset\; F^* \cdot (E_0 + 2,\, E_1 + 2) \;\subset\; \cdots$$

of $k = 2$ plus-seeds, where E_i is the value of $\tilde{\epsilon}$ at the i-th step.

9.2 The Fermat chain

The Fermat partial products $F^{(s)} = \prod_{i=0}^{s-1}(2^{2^i} + 1)$ form a linear chain of plus-seeds for $k = 2$, since the telescoping identity

$$2\prod_{i=0}^{s-1} 2^{2^i} = 2^{2^s} = \prod_{i=0}^{s-1}(2^{2^i} + 1) + 1$$

gives $2\,\tilde{\varphi}^*(F^{(s)}) = \tilde{\epsilon}(F^{(s)}) + 1$. Each seed generates Lehmer factorizations by factoring $\Delta_+(2^{2^s} - 1) = 2^{2^{s+1}} - 2^{2^s} - 1$.

This linear chain is precisely the chain that Theorem 22 produces from the root (3): $E^{(s)} = F_s - 2 = 2^{2^s} - 1$ (by telescoping), so $E^{(s)} + 2 = 2^{2^s} + 1 = F_s$, and the canonical chain extension yields the next Fermat number.

Corollary 25 (Fermat chain identification)**.** *The chain extending* (3) *via Theorem 22 is the sequence of Fermat partial products $F^{(s)} = (F_0, F_1, \ldots, F_{s-1})$ for $s \geq 1$.*

9.3 The first sporadic chain

The Fermat chain is not the only one. Our $|F| \leq 6$ census exhibits a sporadic plus-seed at length 4:

$$S^{(4)} = (5, 5, 5, 43).$$

By Theorem 22 this extends to an infinite chain

$$S^{(4)},\; S^{(5)} = (5, 5, 5, 43, 5377),\; S^{(6)} = (5, 5, 5, 43, 5377, 28901377),\; \ldots$$

where $5375 + 2 = 5377$ and $5375 \cdot 5377 + 2 = 28901377$, etc.

The closed-form enumeration of length-$(s+2)$ extensions extends from the Fermat case to every plus-seed:

Lemma 26 (Coprimality lemma for plus-seeds)**.** *For any $k = 2$ plus-seed $F^* = (x_1, \ldots, x_s)$ with $E = \tilde{\epsilon}(F^*)$ and $M_{F^*} := E^2 + E - 1$, we have $\gcd(M_{F^*}, x_i) = 1$ for every $i \in \{1, \ldots, s\}$.*

Proof. $E = \prod_j x_j$ contains x_i as a factor, so $E \equiv 0 \pmod{x_i}$. Hence $M_{F^*} = E^2 + E - 1 \equiv -1 \pmod{x_i}$, and $\gcd(M_{F^*}, x_i) \mid \gcd(-1, x_i) = 1$. $\square$

Theorem 27 (Plus-seed closedness at $r = s + 2$)**.** *Let F^* be a $k = 2$ plus-seed of length s, with $E = \tilde{\epsilon}(F^*)$ and $M_{F^*} = E^2 + E - 1$. The $k = 2$-Lehmer factorizations of length $s + 2$ extending F^* are in bijection with the divisors $d \leq \sqrt{M_{F^*}}$ of M_{F^*}. The factorization corresponding to d is*

$$\big(F^*,\; d + (E+1),\; M_{F^*}/d + (E+1)\big),$$

and the count is exactly $\tau(M_{F^*})/2$.

Proof. By the $s = r - 2$ finishing-feasibility identity (Proposition 14) applied to F^*, the length-$(s+2)$ extensions are in bijection with divisor pairs $(d, M_{F^*}/d)$ of $M_{F^*} = B \cdot E - A = (E+1) \cdot E - 1$, with $a = d + B$ and $b = M_{F^*}/d + B$ where $B = 2P = E + 1$ for any plus-seed. Integrality follows from $A = 1$. The Lehmer congruence (i.e. $x_i \nmid (a-1)$ and $x_i \nmid (b-1)$ for prior factors x_i) holds because $a - 1 = d + E$ and $E \equiv 0 \pmod{x_i}$ implies $a - 1 \equiv d \pmod{x_i}$, and by Lemma 26, $x_i \nmid d$ (since $d \mid M_{F^*}$ and $\gcd(M_{F^*}, x_i) = 1$). The same applies to $b - 1$. Parity holds because E is a product of odd factors, hence odd, so $E^2 + E$ is even and $M_{F^*} = E^2 + E - 1$ is odd; all its divisors are odd. The remaining ordering and $a \mid b - 1$ checks are as in the Fermat case (where the same argument was originally given for the Fermat specialization). □

The theorem is verified empirically by the $r \leq 7$ census: for each of the eight $k = 2$ plus-seeds with length ≤ 5, the predicted count $\tau(M_{F^*})/2$ matches the number of length-$(s + 2)$ extensions found by exhaustive search:

$(3, 5, 17, 257)$	$\tau/2 = 4$	✓
$(5, 5, 5, 43)$	$\tau/2 = 8$	✓
$(3, 5, 17, 257, 65537)$	$\tau/2 = 16$	✓
$(3, 5, 17, 353, 929)$	$\tau/2 = 4$	✓
$(3, 9, 9, 23, 131)$	$\tau/2 = 1$	✓
$(5, 5, 5, 43, 5377)$	$\tau/2 = 1$	✓
$(5, 5, 5, 47, 453)$	$\tau/2 = 8$	✓
$(5, 7, 7, 7, 133)$	$\tau/2 = 4$	✓

9.4 Fresh plus-seeds and the chain forest

We say a plus-seed of length $s \geq 2$ is a *descendant* if its length-$(s - 1)$ prefix is itself a plus-seed; otherwise it is *fresh*. By Theorem 22, each fresh plus-seed initiates a unique infinite chain of descendant plus-seeds. The set of all $k = 2$ plus-seeds, organized under the parent-child relation $F^* \mapsto (F^*, E + 2)$, forms a disjoint union of infinite linear chains rooted at the fresh plus-seeds.

Table 3: Plus-seed structure by length, $k = 2$, odd spoof. Each fresh plus-seed initiates an infinite descendant chain by Corollary 24; descendant plus-seeds are not listed.

length s	total plus-seeds	fresh	fresh plus-seeds at this length
1	1	1	(3)
2	1	0	—
3	1	0	—
4	2	1	$(5, 5, 5, 43)$
5	6	4	$(3, 5, 17, 353, 929)$, $(3, 9, 9, 23, 131)$, $(5, 5, 5, 47, 453)$, $(5, 7, 7, 7, 133)$

Remark 28 (Two children per plus-seed). A length-s plus-seed F^* produces two distinct length-$(s + 1)$ descendants:

- the *Lehmer descendant* (F^*, E), satisfying $2P(E - 1) = E \cdot E - 1$, which is a $k = 2$-Lehmer factorization by Proposition 14 applied at the plus-seed; and
- the *plus-seed descendant* $(F^*, E + 2)$ from Theorem 22.

The Lehmer descendant is the bottom of an extension tree (no further extension); the plus-seed descendant continues the chain. This dual structure—"one length-$(s+1)$ Lehmer companion plus one length-$(s+1)$ plus-seed continuation"—governs the local structure of $\mathcal{G}_2$ at every plus-seed.

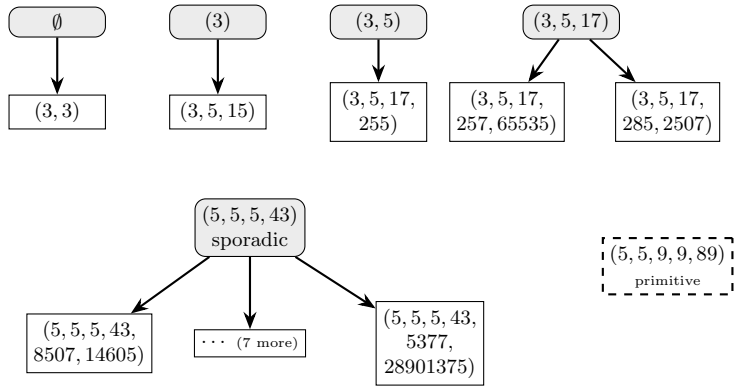

Figure 1: A fragment of $\mathcal{G}_2$. Rounded: plus-seeds. Solid: derived Lehmer factorizations. Dashed: primitive. The $(5,5,5,43)$ subtree exhibits both children: the Lehmer descendant $(5,5,5,43,5375)$ via $x = E$, and the chain extension $(5,5,5,43,5377)$ via $x = E+2$.

10 Partial results at $r = 8$

A complete $r = 8$ census remains out of reach with the current algorithm, even with the finishing-feasibility sharpening of Section 6: the dominant cost is the search through non-plus-seed length-6 prefixes, which is roughly three orders of magnitude more expensive than the corresponding $r = 7$ search. However, the chain theory of Section 9 furnishes *closed-form* enumerations of two canonical $r = 8$ subtrees, providing the first concrete $r = 8$ data.

10.1 The Fermat-6 subtree

By Theorem 27 applied to the Fermat-6 plus-seed $F^{(6)} = (3,5,17,257,65537,4294967297)$, the length-8 extensions are in bijection with divisor pairs of $M_{F^{(6)}} = E^2 + E - 1 = 2^{128} - 2^{64} - 1$. This integer factors as

$$2^{128} - 2^{64} - 1 \;=\; 29 \cdot 101 \cdot 1061 \cdot 1741 \cdot p_{29},$$

where $p_{29} = 62893522383172982457806872591$ is a 29-digit prime, giving $\tau(M_{F^{(6)}})/2 = 16$ length-8 Lehmer extensions. The factorization completes in milliseconds via Pollard-rho ECM, so the entire Fermat-6 subtree at $r = 8$ enumerates in well under one second.

10.2 The sporadic-6 subtree

The sporadic chain rooted at $(5,5,5,43)$ reaches length 6 at $S^{(6)} = (5,5,5,43,5377,28901377)$. By the same theorem, the length-8 extensions of $S^{(6)}$ are governed by the divisors of $M_{S^{(6)}} = E^2 + E - 1$ with $E = 5 \cdot 5 \cdot 5 \cdot 43 \cdot 5377 \cdot 28901377 \approx 8.35 \times 10^{14}$. Direct factorization of $M_{S^{(6)}} \approx 7.0 \times 10^{29}$ is feasible in seconds.

More broadly, every length-6 plus-seed—including the six fresh sporadic length-6 plus-seeds in Table 2—contributes $\tau(M)/2$ length-8 Lehmer extensions in closed form.

10.3 What remains for a complete $r = 8$ census

The closed-form subtrees account for plus-seed length-6 prefixes, and the descent picture (Theorem 4) guarantees that any length-8 Lehmer factorization with descended pair (a, b) has a length-6 plus-seed parent. Therefore the *derived* length-8 factorizations are exactly the union of plus-seed closedness contributions over all length-6 plus-seeds. What remains is to enumerate:

(i) all length-6 $k = 2$-plus-seeds (the chain forest at length 6), and

(ii) all *primitive* length-8 $k = 2$ factorizations.

Both items reduce to a length-6 plus-seed enumeration: by Theorem 22, every length-6 plus-seed either is a chain extension of a length-5 plus-seed (six known) or is *fresh* (a length-5 prefix that is not a plus-seed but extends to a length-6 plus-seed via the closed equation $x_6 = (B + 1)/A$ at the length-5 partial factorization). The fresh enumeration is not yet complete; we leave its determination as the principal computational open problem at $r = 8$.

11 Open questions

1. **Finiteness of primitives.** All known truly-primitive 2-Lehmer factorizations have $|G| \geq 5$. Are there finitely many for each k?

2. **Realizability.** Which spoof factorizations correspond to actual integers $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with each $x_i = p_i^{\alpha_i}$ a prime power?

3. **Fresh plus-seed growth.** At lengths $4, 5, 6$ we observe $1, 4, 6$ fresh plus-seeds respectively (the length-6 count via inverse parent inference; the true length-6 count may be higher). Does the number of fresh plus-seeds at length s grow polynomially, exponentially, or remain bounded?

4. **Density of primitives.** Among the 103 $k = 2$ factorizations with $r \leq 7$, 41 are truly primitive ($\approx 40\%$). Does this proportion converge?

5. **Length-6 plus-seed enumeration.** Direct enumeration of all length-6 plus-seeds (without going through the $r = 7$ Lehmer census) would extend the $r = 8$ partial results of Section 10 to a complete derived-factorization enumeration via Theorem 27.

6. **Sharper $u_{\max}$ bound.** Both length-4 $k = 2$ plus-seeds in our census exhibit $u_{\max} \approx 0.6 \cdot E$ (Remark 18), and spot-checks on three length-5 plus-seeds find no productive u above $\sqrt{EB}$, while the provable bound is $u \leq 2E + 1 \approx 2\sqrt{EB}$. We conjecture the correct bound is $u^2 \leq EB$ (equivalently $u \leq \sqrt{E(E+1)}$) for every $k = 2$ plus-seed; a proof would likely require analyzing the divisor structure of $M_u = u^2EB + uC + (EB)^2$ in the residue class $-EB \pmod{u}$ in the regime $u^2 > EB$, which the divisor-density heuristic does not capture.

A Full census listing (organized by chain forest)

The complete $r \leq 7$, $k = 2$ census is presented below in hierarchical form: each plus-seed appears as a section header (annotated with its chain identification, depth in the chain, and $\tilde{\epsilon}$ value), followed by its descended-pair and Lehmer-companion children. The chain ordering is Fermat first (depth 0 to 5), then sporadic chains in the order they first appear in Table 2. Descended children within a

section are sorted by the second-to-last factor x_{r-1}. Truly primitive factorizations (those with no plus-seed parent of either kind) appear at the end, grouped by length.

Table 4: The $r \leq 7$ $k = 2$-Lehmer factorizations grouped by parent plus-seed. Each plus-seed appears as a section header, followed by its descended-pair and Lehmer-companion children. Truly primitive factorizations (no plus-seed parent of either kind) appear at the end.

(3) (Fermat, depth 1, $\tilde{\epsilon} = 3$):	
(3, 5, 15)	descended
(3, 5) (Fermat, depth 2, $\tilde{\epsilon} = 15$):	
(3, 5, 17, 255)	descended
(3, 5, 17) (Fermat, depth 3, $\tilde{\epsilon} = 255$):	
(3, 5, 17, 257, 65535)	descended
(3, 5, 17, 285, 2507)	descended
(3, 5, 17, 257) (Fermat, depth 4, $\tilde{\epsilon} = 65535$):	
(3, 5, 17, 257, 65537, 4294967295)	descended
(3, 5, 17, 257, 65555, 226112997)	descended
(3, 5, 17, 257, 65717, 23794275)	descended
(3, 5, 17, 257, 68975, 1314417)	descended
(3, 5, 17, 257, 65537) (Fermat, depth 5, $\tilde{\epsilon} = 4294967295$):	
(3, 5, 17, 257, 65537, 4294967297, 18446744073709551615)	descended
(3, 5, 17, 257, 65537, 4294967307, 1676976737878111325)	descended
(3, 5, 17, 257, 65537, 4294967367, 259813301047285385)	descended
(3, 5, 17, 257, 65537, 4294967375, 233503093781227857)	descended
(3, 5, 17, 257, 65537, 4294968077, 23619394908814395)	descended
(3, 5, 17, 257, 65537, 4294968165, 21227557884627347)	descended
(3, 5, 17, 257, 65537, 4294968305, 18282208526299887)	descended
(3, 5, 17, 257, 65537, 4294972905, 3288780203224487)	descended
(3, 5, 17, 257, 65537, 4294978395, 1662022861452077)	descended
(3, 5, 17, 257, 65537, 4295028995, 298983922990677)	descended
(3, 5, 17, 257, 65537, 4295038935, 257500129211417)	descended
(3, 5, 17, 257, 65537, 4295047007, 231424601693025)	descended
(3, 5, 17, 257, 65537, 4295755325, 23413007171307)	descended
(3, 5, 17, 257, 65537, 4295844117, 21042504669635)	descended
(3, 5, 17, 257, 65537, 4300626777, 3263735907095)	descended
(3, 5, 17, 257, 65537, 4357221587, 300607780005)	descended
(5, 5, 5, 43) (sporadic root, $\tilde{\epsilon} = 5375$):	
(5, 5, 5, 43, 5375)	companion
(5, 5, 5, 43, 5377, 28901375)	descended
(5, 5, 5, 43, 5387, 2632285)	descended
(5, 5, 5, 43, 5407, 937505)	descended
(5, 5, 5, 43, 5477, 291475)	descended
(5, 5, 5, 43, 5717, 90115)	descended

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(5, 5, 5, 43, 6215, 39817)	descended
(5, 5, 5, 43, 6487, 31385)	descended
(5, 5, 5, 43, 8507, 14605)	descended

(3, 5, 17, 353, 929) (sporadic root, $\tilde{\epsilon} = 83623935$):

(3, 5, 17, 353, 929, 83623935)	companion
(3, 5, 17, 353, 929, 83623937, 6992962672132095)	descended
(3, 5, 17, 353, 929, 83623955, 368050746176997)	descended
(3, 5, 17, 353, 929, 84866225, 5712718767)	descended
(3, 5, 17, 353, 929, 107227427, 379892085)	descended

(3, 9, 9, 23, 131) (sporadic root, $\tilde{\epsilon} = 732159$):

(3, 9, 9, 23, 131, 732159)	companion
(3, 9, 9, 23, 131, 732161, 536058265599)	descended

(5, 5, 5, 43, 5377) (sporadic, depth 1, $\tilde{\epsilon} = 28901375$):

(5, 5, 5, 43, 5377, 28901377, 835289534693375)	descended

(5, 5, 5, 47, 453) (sporadic root, $\tilde{\epsilon} = 2661375$):

(5, 5, 5, 47, 453, 2661375)	companion
(5, 5, 5, 47, 453, 2661377, 7082922213375)	descended
(5, 5, 5, 47, 453, 2661587, 33571000485)	descended
(5, 5, 5, 47, 453, 2662317, 7529674715)	descended
(5, 5, 5, 47, 453, 2665797, 1604769395)	descended
(5, 5, 5, 47, 453, 2669445, 880455347)	descended
(5, 5, 5, 47, 453, 2859927, 38334425)	descended
(5, 5, 5, 47, 453, 3594207, 10254305)	descended
(5, 5, 5, 47, 453, 4363935, 6821537)	descended

(5, 7, 7, 7, 133) (sporadic root, $\tilde{\epsilon} = 228095$):

(5, 7, 7, 7, 133, 228095)	companion
(5, 7, 7, 7, 133, 228097, 52027785215)	descended
(5, 7, 7, 7, 133, 230447, 22358065)	descended
(5, 7, 7, 7, 133, 232717, 11487035)	descended
(5, 7, 7, 7, 133, 232885, 11092067)	descended

(3, 5, 17, 257, 65729, 22318913) (sporadic root, $\tilde{\epsilon} = 96139834027933695$):

(3, 5, 17, 257, 65729, 22318913, 96139834027933695)	companion

(3, 5, 17, 365, 855, 7234467) (sporadic root, $\tilde{\epsilon} = 575712553701375$):

(3, 5, 17, 365, 855, 7234467, 575712553701375)	companion

(3, 11, 11, 11, 677, 3659) (sporadic root, $\tilde{\epsilon} = 9891231999$):

(3, 11, 11, 11, 677, 3659, 9891231999)	companion

(5, 5, 5, 43, 5413, 786349) (sporadic root, $\tilde{\epsilon} = 22878725861375$):

(5, 5, 5, 43, 5413, 786349, 22878725861375)	companion

(5, 5, 5, 53, 215, 158265) (sporadic root, $\tilde{\epsilon} = 225428709375$):

(5, 5, 5, 53, 215, 158265, 225428709375)	companion

(5, 7, 7, 13, 13, 619) (sporadic root, $\tilde{\epsilon} = 25629695$):

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(5, 7, 7, 13, 13, 619, 25629695)	companion

Truly primitive factorizations (no plus-seed parent):

(5, 5, 9, 9, 89)	$r = 5$
(3, 5, 17, 377, 1217, 2295)	$r = 6$
(3, 5, 17, 395, 1059, 2303)	$r = 6$
(3, 5, 33, 53, 69, 8343)	$r = 6$
(5, 5, 5, 45, 807, 267023)	$r = 6$
(5, 5, 5, 47, 457, 50659)	$r = 6$
(5, 5, 5, 65, 129, 2325)	$r = 6$
(3, 5, 17, 257, 65543, 613622973, 8815156799949437)	$r = 7$
(3, 5, 17, 257, 65543, 613627757, 78010845360333)	$r = 7$
(3, 5, 17, 257, 65555, 226113003, 8596590110981999)	$r = 7$
(3, 5, 17, 257, 65583, 91446429, 1862725578762405)	$r = 7$
(3, 5, 17, 257, 65855, 14660375, 175338855)	$r = 7$
(3, 5, 17, 257, 66023, 8884637, 90028574052093)	$r = 7$
(3, 5, 17, 257, 66035, 12360093, 29069129)	$r = 7$
(3, 5, 17, 257, 66527, 4399443, 249102573029243)	$r = 7$
(3, 5, 17, 257, 70953, 1015619, 5544845)	$r = 7$
(3, 5, 17, 257, 72113, 719417, 599253543)	$r = 7$
(3, 5, 17, 257, 72939, 645693, 6040018747905)	$r = 7$
(3, 5, 17, 257, 75273, 506627, 1912164594413)	$r = 7$
(3, 5, 17, 257, 85185, 288275, 19698725)	$r = 7$
(3, 5, 17, 257, 103943, 199577, 1593393)	$r = 7$
(3, 5, 17, 257, 104093, 176927, 1175218174293)	$r = 7$
(3, 5, 17, 263, 9585, 27948617, 125635487981969)	$r = 7$
(3, 5, 17, 263, 11159, 78249, 506759)	$r = 7$
(3, 5, 17, 269, 5279, 19059029, 809372846673)	$r = 7$
(3, 5, 17, 269, 5283, 5104049, 80419259566169)	$r = 7$
(3, 5, 17, 287, 2363, 4762259, 252056385)	$r = 7$
(3, 5, 17, 363, 887, 36705, 1780076619)	$r = 7$
(3, 5, 17, 387, 755, 837215, 12806102345)	$r = 7$
(3, 5, 17, 419, 659, 172587, 68725637)	$r = 7$
(3, 5, 23, 95, 207, 215, 85803023)	$r = 7$
(3, 5, 27, 45, 233, 42489, 1389617)	$r = 7$
(3, 5, 33, 33, 359, 4869, 2502245)	$r = 7$
(5, 5, 5, 43, 5479, 327773, 2239165)	$r = 7$
(5, 5, 5, 43, 5603, 132673, 9536030197)	$r = 7$
(5, 5, 5, 43, 5623, 122489, 119765185)	$r = 7$
(5, 5, 5, 43, 5759, 80825, 2587288073)	$r = 7$
(5, 5, 5, 43, 6985, 23335, 28261225505)	$r = 7$
(5, 5, 5, 43, 8749, 13943, 1335400765)	$r = 7$
(5, 5, 5, 47, 535, 2947, 36041983)	$r = 7$
(5, 5, 5, 49, 355, 3619, 183001469)	$r = 7$

References

[1] E. Hasanalizade, *Some remarks on a paper of Molnar and Singh*, Integers **26** (2026).

[2] D. H. Lehmer, *On Euler's totient function*, Bull. Amer. Math. Soc. **38** (1932), 745–751.

[3] G. Molnar and G. Singh, *Spoof Lehmer factorizations*, Integers **26** (2026), #A33.

[4] P. Nielsen, *Odd perfect numbers, Diophantine equations, and upper bounds*, Math. Comp. **84** (2015), 2549–2567.

[5] A. Meurer et al., *SymPy: symbolic computing in Python*, PeerJ Computer Science **3** (2017), e103.